Selected Topics in HTML5 and CSS3: SEO

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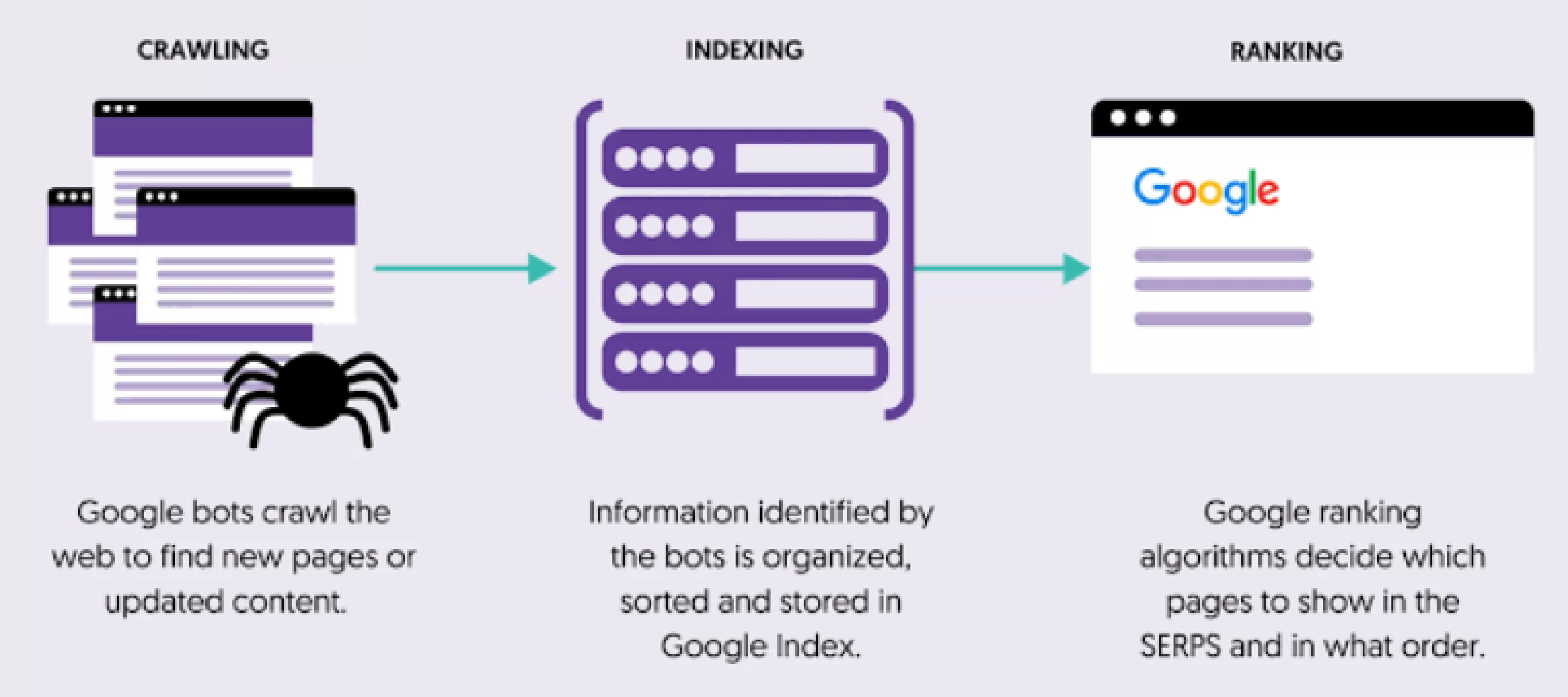
How search engines work?



Search Engine Optimization

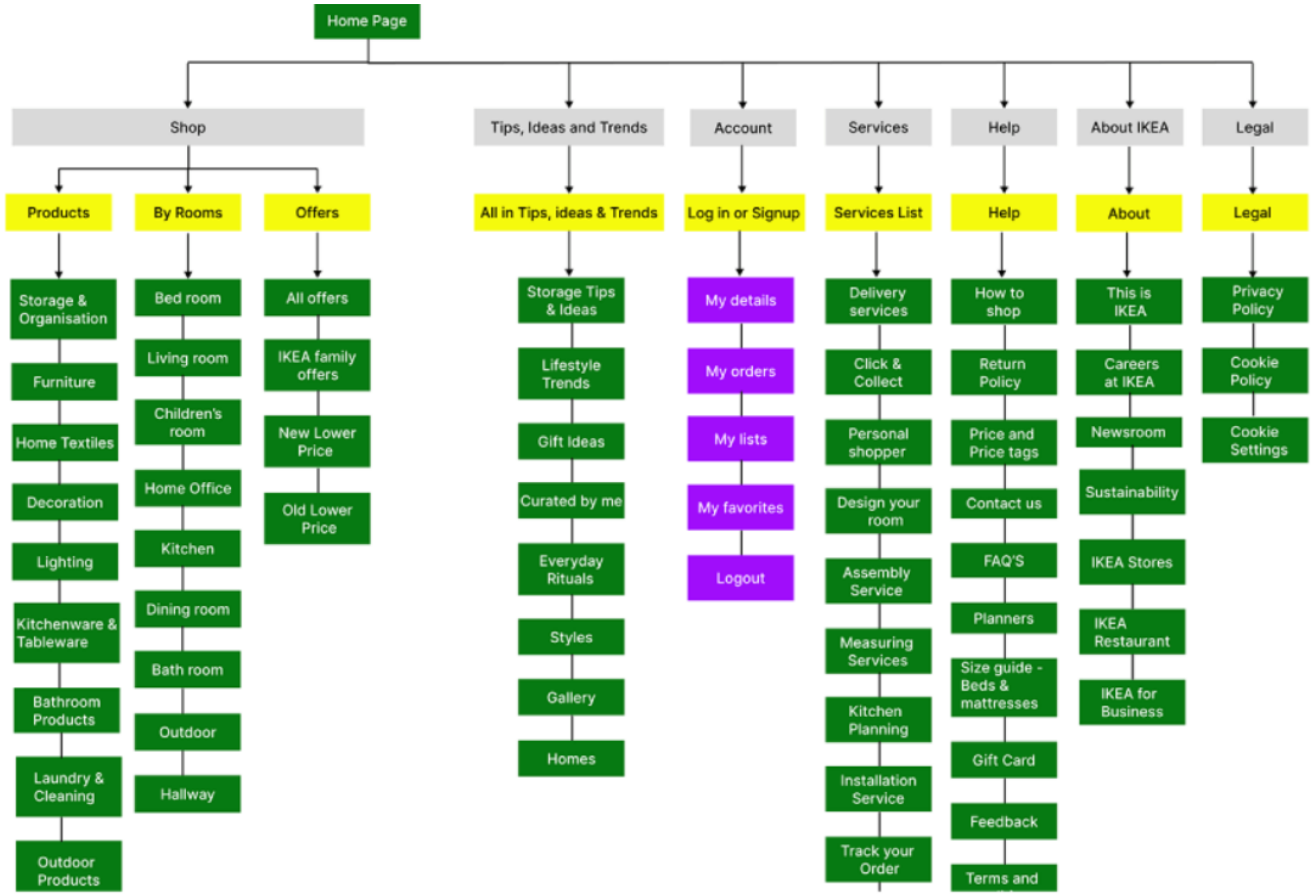
- SEO is the practice of improving your website's content, structure, and visibility to rank higher on **organic** search engine results for relevant searches.
- Types of web traffic:
  - Direct traffic: Users enter the website URL directly.
  - Organic search traffic: Users reach the website through unpaid search-engine results.
  - Paid search traffic: Users arrive by clicking paid advertisements displayed in search results (Google Ads campaign).
  - Referral traffic: Users reach a website by clicking a link on another website. A news website links to a university website.
- Why SEO Matters: Organic search drives ~50% of all web traffic.
  - Higher visibility = more **domain authority**, trust, and clicks.
- How can you improve the three steps with respect to your site: crawling, indexing and ranking?

Improve crawling of your website

- Use an organized site structure with **high internal linking**
- Create an **XML sitemap** to guide search engine crawlers
- Regularly update content to signal freshness
- Avoid broken links and fix crawl errors in Google Search Console

Sitemap

- Sitemaps are an easy way for webmasters to inform search engines about pages on their sites that are available for crawling.



Sitemap

- Web crawlers usually discover pages from links within the site and from other sites.
  - Sitemaps supplement this data to allow crawlers to pick up all URLs in the Sitemap and learn about those URLs using the associated metadata especially for sites with complex structures or limited internal linking.
- There are two types of sitemaps:
  - XML and HTML
- An HTML sitemap is like a master list of all your important website pages, presented as an easy-to-read webpage.
  - User-Friendly Navigation for both search engines and people
- Example: https://www.dau.ac.in/sitemap

XML Sitemap

- XML Sitemap is an XML file that lists URLs for a site along with additional metadata about each URL so that search engines can more intelligently crawl the site.
  - when it was last updated,
  - how often it usually changes, and
  - how important it is, relative to other URLs in the site

Example: https://www.dau.ac.in/sitemap.xml

XML Sitemap Protocol

- XML file adhering to the sitemap protocol.
- Has URL and add optional metadata tags like <lastmod>, <changefreq>, and <priority>.
- The Sitemap protocol format consists of XML tags. All data values in a Sitemap must be entity-escaped. The file itself must be UTF-8 encoded.
- The Sitemap must:
  - Begin with an opening <urlset> tag and end with a closing </urlset> tag.
  - Specify the namespace (protocol standard) within the <urlset> tag.
  - Include a <url> entry for each URL, as a parent XML tag.
  - Include a <loc> child entry for each <url> parent tag.
- All other tags are optional. Support for these optional tags may vary among search engines.
- More details

XML sitemap example

```xml
<?xml version="1.0" encoding="UTF-8"?>
  <urlset xmlns="http://www.sitemaps.org/schemas/sitemap/0.9">
       <url>
              <loc>http://www.example.com/</loc>
              <lastmod>2005-01-01</lastmod>
              <changefreq>monthly</changefreq>
              <priority>0.8</priority>
       </url>
  </urlset>
```

How to create sitemaps

- Manual Creation
  - Write XML using proper tags (<urlset>, <url>, <loc>, etc.)
  - For HTML sitemaps, use structured lists with links to all pages.
- Using CMS Plugins
  - WordPress: Yoast SEO, Rank Math.
  - Joomla, Drupal: Built-in or plugin-based sitemap generators.
- Online Sitemap Generators
  - Example: XML-Sitemaps.com, Screaming Frog SEO Spider.
- Command Line & AutomationPython tools:
  - python-sitemap, django-sitemap.

Validating Sitemaps

- Google Search Console
  - Submit and check for errors.
- Online Validators
  - Example: https://www.xml-sitemaps.com/validate-xml-sitemap.html.
- Schema Validation
  - Ensure compliance with sitemap protocol (https://www.sitemaps.org).

Steps in using the sitemap

- Create the sitemap
- place it on your web server
- Inform the search engines of its location:
  - Submit via Google Search console: Log in, select your property, click Sitemaps under the Indexing menu, paste your full sitemap URL and click Submit.
  - Specify in your site's robots.txt eg.
    - Add in .txt: Sitemap: http://www.example.com/sitemap.xml
  - Submit sitemap using an HTTP request
    - <searchengine_URL>/ping?sitemap=sitemap_url
- Search engines then retrieve your sitemap and make the URLs available to their crawlers

Robots.txt

- robots.txt is a text file placed at the root of a website to communicate with web crawlers and bots about which parts of the site they are allowed or disallowed to access – aka *robot exclusion protocol*.
- There are two important considerations when using /robots.txt:
  - robots.txt is a guideline, not an enforcement - malicious bots may ignore it.
  - the /robots.txt file is a publicly available file. Anyone can see what sections of your server you don't want robots to use.
- It does not hide content from search results if other pages link to it
  - use noindex meta tag or password protection for that.
- Example: https://www.dau.ac.in/robots.txt

Example robots.txt files

```
User-agent: *
Disallow: /
```

The "User-agent: *" means this section applies to all robots. The "Disallow: /" tells the robot that it should not visit any pages on the site.

```
User-agent: *
Disallow:
```

To allow all robots complete access

```
User-agent: Google
Disallow:

User-agent: *
Disallow: /
```

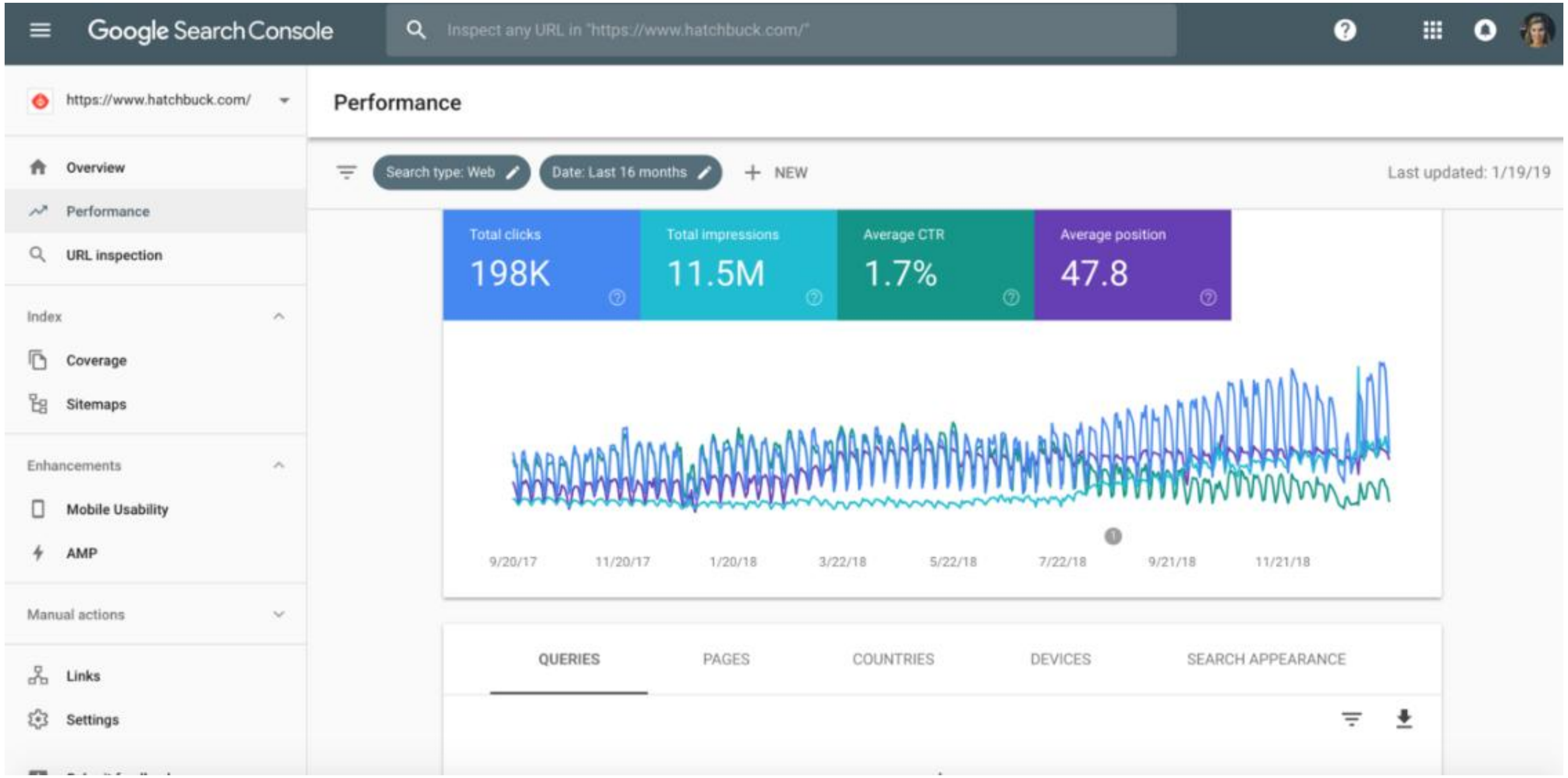
To allow a single robot and not allow others

The metrics

- Clicks: The number of times a user clicked your website link in the Google search results page.
- Impressions: The number of times your website link appeared on a Google search results page for a user.
- CTR (Click-Through Rate): The percentage of impressions that turned into clicks (calculated as Clicks divided by Impressions).
- Average Position: Where your site usually ranks on Google. Position 1 is the top of the first page, 10 is the bottom of the first page, and anything over 10 is usually on page two or deeper.
- For example:
  - A higher CTR means?
  - your title and description look helpful and attractive.

Google Search Console

- Google Search Console shows how Google (search engine) sees your site and how users interact with it in search results.



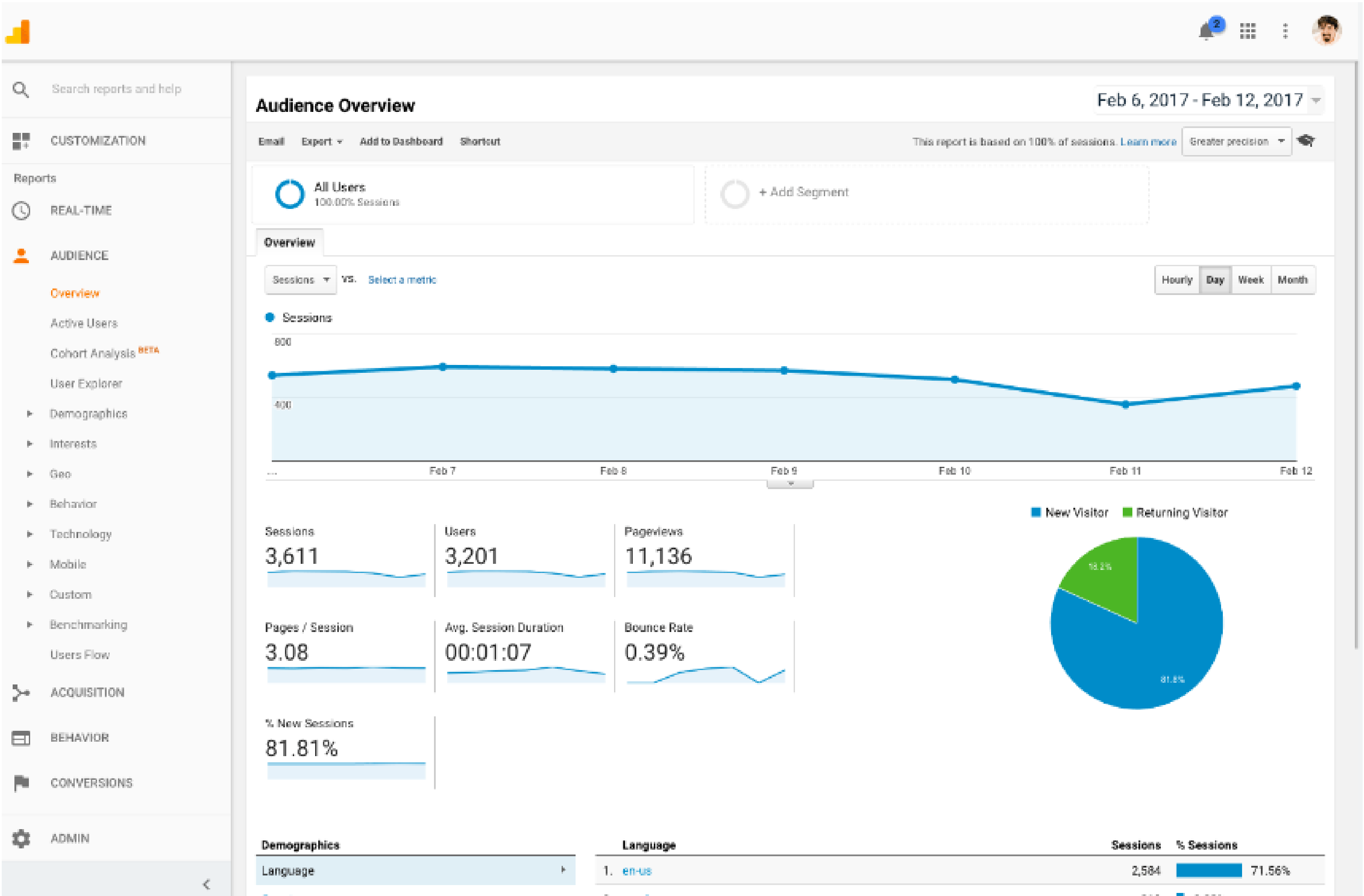
Google Search Console

- Fixing technical bugs: Identify broken links, indexing errors, or mobile display issues.
- Optimizing keywords: See the exact phrases people type into Google to find your pages.
- Submitting sitemaps: Tell Google directly which pages you want crawled and indexed.
- Monitoring security: Get alerts for malware, hacking, or manual spam penalties.

Metrics

- Acquisition: total users
- Engagement: engagement time
- Monetization: conversions
- Retention: returning users

Google Analytics



Google Analytics

- Analyzing traffic sources: See if your visitors come from Google, Facebook, emails, or direct typing.
- Tracking user behavior: Learn which pages people visit most, how long they stay, and where they leave.
- Measuring conversions: Track online sales, contact form sign-ups, or PDF downloads.
- Understanding audience: See the geographic location, language, and device types of your visitors.

Types of SEO

- On page SEO: content, keywords, HTML tags
- Off-page SEO: backlinks, social signals
- Technical SEO: site speed, mobile-friendliness

On-Page SEO

- High quality Content and Keywords
  - Use main keywords: Place your focus keyword naturally in the title (H1), introduction, and a few subheadings to effectively organize the content.
- Target **search intent**: Match content with what the user wants to find, whether it is a guide, a product, or a local service. Types of Intent:
  - Informational (“What is SEO?”)
  - Navigational (“Twitter login”)
  - Transactional (“Buy running shoes online”)
- Use keyword planner tools such as Semrush to perform **keyword research**
  - Identify seed keywords based on your core topics or services and use keyword tools for insights
  - Different keywords may be more suitable for different intents
  - Well performing and competitor keywords
  - Target long-tail keywords with lower competition and higher relevance
  - Analyze competitor keywords to find gaps and opportunities
- Tool: Semrush

On-Page SEO

- **Keyword difficulty (KD)**: metric that estimates how hard it is to rank on the first page of search engine results for a specific search term. It is typically scored on a scale from 0 to 100, where higher numbers mean tougher competition.
- Example:
  - Competitive: "web hosting" (Hard to rank for)
  - Easier (Long-tail): "best web hosting for small business websites" (Easier to rank for)
  - Competitive: "dentist"
  - Easier (Local): "emergency dentist in Ahmedabad" :

Off-Page SEO and Link Building

- Off-page SEO refers to all activities performed outside your website to improve its reputation and trust across the web and hence, rankings in search engines.
- Techniques
  - backlinks from high-domain authority domains .
  - Sharing content of social media platforms
  - guest blogging

Technical SEO

- Sitemaps
- Robots.txt
- Structured data
- Core web vitals

Structured Data

- Data organized in a predictable format for easy interpretation by machines.
- Enhances search engines' understanding of web content.

- Why is it important?
  - Enables rich search results (snippets, carousels, knowledge panels).
  - Improves SEO visibility and indexing.

Vocabularies

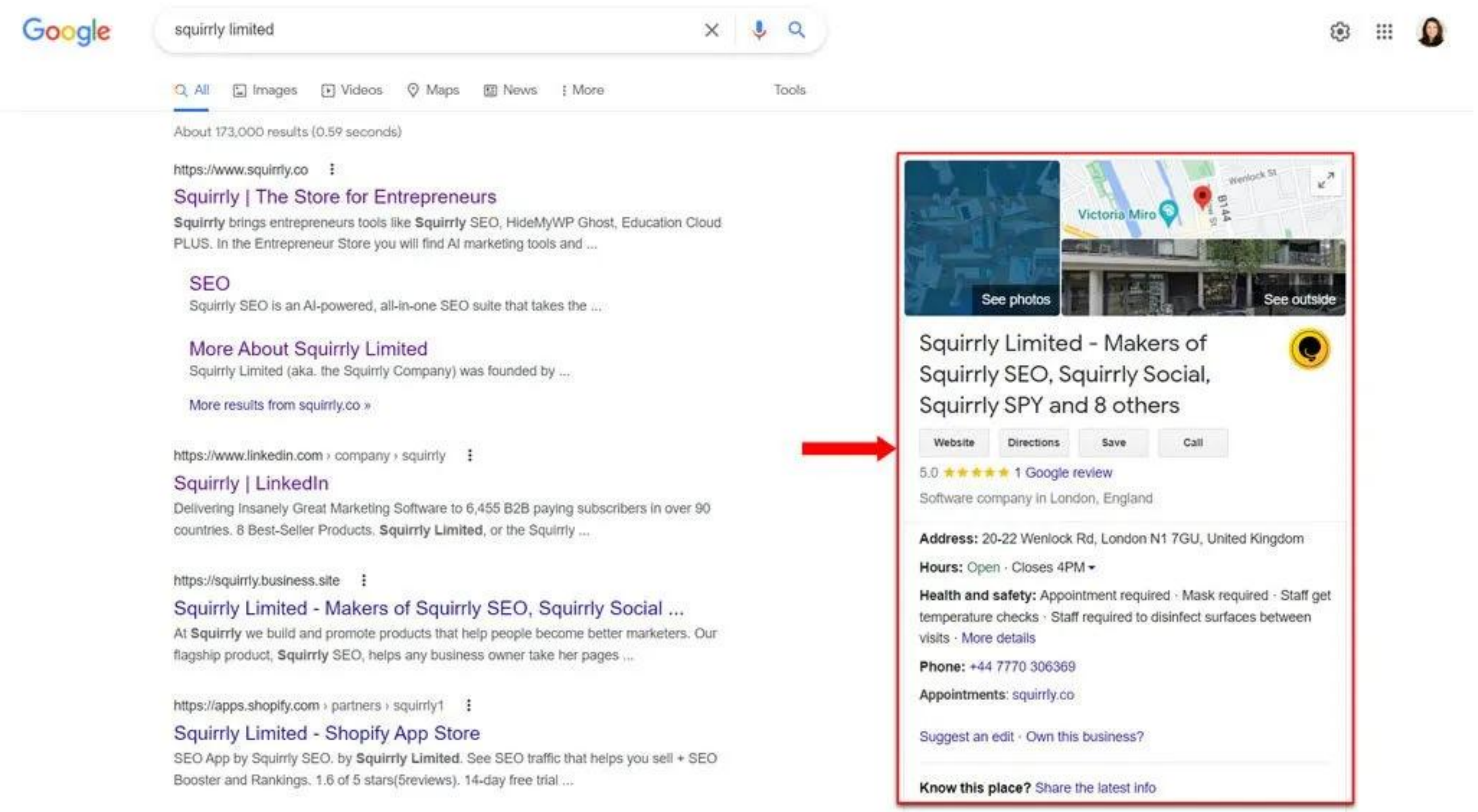
- Vocabulary: Set of standard terms to describe things (people, products, events).
- Schema.org:
  - A collaborative vocabulary supported by Google, Microsoft, Yahoo, and Yandex.
  - Provides schemas (types, properties) to mark up content.
  - Used with microdata, RDFa, or JSON-LD.
- These vocabularies cover entities, relationships between entities and actions
- semantic information in the content can be conveyed in a way search engines can understand
- Structured data testing tool/ Rich results test available

JSON for linking data (JSON-LD)

- JSON-LD is a scripting language that allows publishers to communicate important information to search engines.
- creates a network of standards-based, machine-readable data across Websites.
- JSON injects metadata in the head section. This is used as a data block added as a script element. No need to tweak actual HTML data.
- Loads faster, better indexing in search engines
  It is the choice of search engines like Google and Yahoo
- JSON-LD structured data influences how Google will create rich snippets for your URLs.

Rich snippet

- Rich snippets are enhanced search results shown on search engines
- Generated when a webpage includes structured data markup (like JSON-LD, Microdata, or RDFa) that helps search engines understand the page content better.
- Importance:
  - Make your listing stand out in search results
  - Can improve click-through rates (CTR).


Google
squirrly limited
All
Images
Videos
Maps
News
More
Tools
About 173,000 results (0.59 seconds)
https://www.squirrly.co
Squirrly | The Store for Entrepreneurs
Squirrly brings entrepreneurs tools like Squirrly SEO, HideMyWP Ghost, Education Cloud PLUS. In the Entrepreneur Store you will find AI marketing tools and ...
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SEO App by Squirrly SEO. by Squirrly Limited. See SEO traffic that helps you sell + SEO Booster and Rankings. 1.6 of 5 stars(5reviews). 14-day free trial ...
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B144
See photos
See outside
Squirrly Limited - Makers of Squirrly SEO, Squirrly Social, Squirrly SPY and 8 others
Website
Directions
Save
Call
5.0 ★★★★★ 1 Google review
Software company in London, England
Address: 20-22 Wenlock Rd, London N1 7GU, United Kingdom
Hours: Open · Closes 4PM
Health and safety: Appointment required · Mask required · Staff get temperature checks · Staff required to disinfect surfaces between visits · More details
Phone: +44 7770 306369
Appointments: squirrly.co
Suggest an edit · Own this business?
Know this place? Share the latest info

Example

```html
 <div>
    <h1> Avatar </h1>
    <div>
        Director: James Cameron (born August 16, 1954)
    </div>
    <span>Science fiction</span>
    <a href="../movies/avatar-theatrical-trailer.html">Trailer</a>
</div>
```

Without semantic markup

```html
<script type="application/ld+json">
{   "@context":"http://schema.org/",
    "@type": "Movie",
    "name": "Avatar",
    "director":
    {   "@type": "Person",
        "name": "James Cameron",
        "birthDate": "1954-08-16"
    },
    "genre": "Science fiction",
    "trailer": "../movies/avatar-theatrical-trailer.html"
}
</script>
```

JSON-LD

```
<script type="application/ld+json">
{
  "@context": "https://schema.org",
  "@type": "Restaurant",
  "name": "Delicious Bites",
  "image":
"https://example.com/images/logo.png",
  "@id":
"https://deliciousbites.example.com",
  "url":
"https://deliciousbites.example.com",
  "telephone": "+1-555-555-1234",
  "address": {
    "@type": "PostalAddress",
    "streetAddress": "123 Foodie Lane",
    "addressLocality": "Gourmet City",
    "addressRegion": "CA",
    "postalCode": "90001",
    "addressCountry": "US"
  },
  "geo": {
    "@type": "GeoCoordinates",
    "latitude": 34.0522,
    "longitude": -118.2437
  },
"openingHoursSpecification": [
    {
      "@type": "OpeningHoursSpecification",
      "dayOfWeek": [
        "Monday",
        "Tuesday",
        "Wednesday",
        "Thursday",
        "Friday"
      ],
      "opens": "09:00",
      "closes": "22:00"
    },
    {
      "@type": "OpeningHoursSpecification",
      "dayOfWeek": "Saturday",
      "opens": "10:00",
      "closes": "23:00"
    }
  ],
  "servesCuisine": ["Italian",
"Continental"],
  "priceRange": "$$",
  "menu":
"https://deliciousbites.example.com/menu",
  "acceptsReservations": "True"
}
</script>
```

Validation

- Rich results test : https://search.google.com/test/rich-results

Core Web Vitals

- **Core Web Vitals (CWV)** are a set of metrics developed by Google to measure the **real-world user experience** of a webpage. They answer three important questions:
  - Does the page load quickly?
  - Does the page respond quickly when the user interacts?
  - Does the page remain visually stable while loading

- **Metrics**
  - Largest Contentful Paint (LCP) measures loading performance: Measures the time it takes for the largest content element on a webpage to load. A good LCP score is under 2.5 seconds.
  - Interaction to Next Paint (INP) measures responsiveness: Measures the time it takes for a webpage to respond to a user's interaction (like a click or tap). A good INP score is 200 milliseconds or less.
  - Cumulative Layout Shift (CLS) measures visual stability: how much unexpected layout shifts occur while it's loading. A good CLS score is 0.1 or less.

- **Based** on these vitals, webpages are classified as:
  - Good
  - Needs Improvement
  - Poor